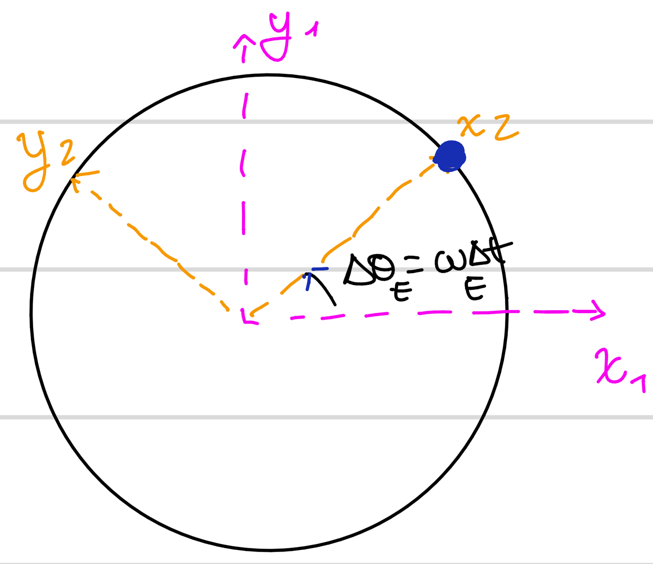
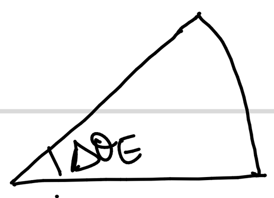
Caso guiado - 500 km



$$S_1 \xrightarrow[\text{}]{+\Delta\theta_E \hat{k}_{1,2}} S_2$$

ECI — inercial; ECEF — no inercial; $\dot{\Delta\theta}_E \hat{k}_{1,2} = \omega_E \hat{k}_{1,2}$

$$\begin{bmatrix} \hat{i}_1 \\ \hat{j}_1 \\ \hat{k}_1 \end{bmatrix} = Q_{12} \begin{bmatrix} \hat{i}_2 \\ \hat{j}_2 \\ \hat{k}_2 \end{bmatrix} \rightsquigarrow \begin{cases} \hat{i}_1 = \cos\omega_E\Delta t\, \hat{i}_2 - \cos\omega_E\Delta t\, \hat{j}_2 \\ \hat{j}_1 = \operatorname{sen}\omega_E\Delta t\, \hat{i}_2 + \cos\omega_E\Delta t\, \hat{j}_2 \end{cases}$$



$$\omega = \frac{d\theta}{dt} \rightarrow d\theta_E = \omega_E dt$$

h (km)	r [km]	v (km/s)	T (min)	N (órbitas/día)
400	6771	7'673	92'41	15'58
500	6871	7'617	94'47	15'24
600	6971	7'562	96'54	14'92

$$\begin{cases} r = r_E + h \\ v = \sqrt{\frac{\mu}{r}} = \sqrt{\frac{3'986 \cdot 10^{14}}{r\,[\text{km}]}} \\ T = 2\pi\sqrt{\frac{r^3}{\mu}} \\ N = \frac{24}{(T/60)} \quad ; T\,[\text{min}] \end{cases}$$

(suponiendo órbita circular)